UCL AGENTVERSE 2026 | ROBOTICS TRACK | NVIDIA & GOOGLE DEEPMIND

Robopod

A hands-free camera assistant for solo creators

The problem

Solo creators and educators who film themselves face a trade-off. A handheld camera pulls attention away from the work, and a fixed tripod limits every video to a single static angle.

Robopod is built to take over camera movement so creators keep their hands free and stay focused on the content.

Key features

Robopod turns a 6-DoF arm into a hands-free camera assistant in MuJoCo, and plans, runs and checks every shot.

- **Room from a photo:** one image becomes a collision model, mapped by an 18-pose survey.
- **Shots from plain language:** a brief becomes a validated multi-shot plan from local Gemma 4.
- **Live tracking:** orbit, dolly in, low angle and tracking shots follow you.
- **Self-correcting framing:** loss fell from **1.506 to 0.309** on a presenter shot.
- **Built-in coach:** flags drift beyond 0.15 m or a slumping posture.

How it fits together

1 Reconstruct

Gemma 4 reads the room photo and writes collision geometry. An 18-pose survey becomes a hashed scene map.

2 Plan

Gemma 4 turns your brief and survey views into a shot plan, schema-checked before anything moves.

3 Control

A 200 Hz loop blends your live pose with the shot offset, aims the lens at you, and solves IK with mink.

4 Verify

Framing loss and a digital twin score the result, shown live in the local studio app.

Where Gemma runs, and how fast

Gemma runs locally through Ollama (`gemma4:e2b`) or as `gemma-4-31b-it`, on an Apple M-series chip, an NVIDIA RTX card or a plain CPU, with a rule-based fallback. Every model runs on the local machine, so there are no cloud calls at any stage.

Job	Runs on	Latency
Room to collision XML (Gemma 4)	Local edge	One-time setup
Shot parsing and planning (Gemma 4)	Local edge	Under 120 ms, before each shot
Pose tracking (MediaPipe)	Local CPU or NPU	30 Hz
Path blend and IK (mink, MuJoCo)	Local CPU or GPU	200 Hz, under 3.5 ms per step
Framing check and digital twin	Local CPU	10 Hz check, twin under 0.15 ms

Critical failure modes and safety

The arm hits a limit or obstacle. Joint limits are checked every 0.002 s step and the run halts on a breach. Targets clamp to a 0.75 m reach, below the 0.85 m singularity, with a 6 cm sag margin. The solver fails over from daqp to quadprog.

A shot can't be framed. An unsatisfied shot is never reported as done. Up to three teleprompter rounds follow, then an explicit request to reposition the subject or product.

Stale or failed AI output. The scene map hash is verified before any move, and Restart cancels in-flight plans. If Gemma fails or returns bad output, a fallback room appears with a red floor mat. Servers are loopback only.

Honest limits. A fixed-base UR5e can't match a dolly track and the lens has no optical zoom. One photo yields simple boxes and cylinders, and the virtual human is pinned without foot contact physics. **39 of 39 tests pass.**